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shreyas@Grigori:~$ nano process_states.c
shreyas@Grigori:~$ gcc process_states.c -o process_states
shreyas@Grigori:~$ ./process_states
=== Process State Demonstration ===
[Initial Parent] PID: 2397, PPID: 361 | State:  R (running)

-- Child Process Started --
[Child Running] PID: 2398, PPID: 2397 | State:  R (running)
Child entering sleeping/waiting state (sleep 3s)...
[Parent Observing Child (Sleeping)] PID: 2398, PPID: 361 | State:          S (sleeping)
[Child Resumed] PID: 2398, PPID: 2397 | State:  R (running)
--- Child Exiting ---

--- Child Terminated ---
[Parent After wait()] PID: 2397, PPID: 361 | State:          R (running)
shreyas@Grigori:~$ nano task2.c
shreyas@Grigori:~$ gcc task2.c -o task2
shreyas@Grigori:~$ ./task2 &
[1] 2405
shreyas@Grigori:~$ Process started
PID   : 2405
PPID  : 361
Process State: Running
Process State: Sleeping/Waiting
Process State: Running again
Process State: Terminating
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--- Child Terminated ---
[Parent After wait()] PID: 2397, PPID: 361 | State:      R (running)
shreyas@Grigori:~$ nano task2.c
shreyas@Grigori:~$ gcc task2.c -o task2
shreyas@Grigori:~$ ./task2 &
[1] 2405
shreyas@Grigori:~$ Process started
PID   : 2405
PPID  : 361
Process State: Running
Process State: Sleeping/Waiting
Process State: Running again
Process State: Terminating
C
[1]+  Done                  ./task2
shreyas@Grigori:~$ ps -o pid,ppid,stat,cmd -C task2
  PID   PPID STAT CMD
shreyas@Grigori:~$ kill -STOP $(pgrep task2)
kill: usage: kill [-s sigspec | -n signum | -sigspec] pid | jobspec ... or kill -l [sigspec]
shreyas@Grigori:~$ ps -o pid,ppid,stat,cmd -C task2
  PID   PPID STAT CMD
shreyas@Grigori:~$ kill -CONT $(pgrep task2)
kill: usage: kill [-s sigspec | -n signum | -sigspec] pid | jobspec ... or kill -l [sigspec]
shreyas@Grigori:~$ ps -o pid,ppid,stat,cmd -C task2
  PID   PPID STAT CMD
shreyas@Grigori:~$

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